

Exercise 4

Task one will be studied in case it was in the mockup exam.

Parameterized testing was learned no need for further learning

Task 2 – Test Planning and Test Strategy:

In your role as test manager, you are responsible for the test of the application “zoom”:

2.1.) Define test objectives, scope and risks of testing:

Test objectives meaning: what you want to test and verify is working. Verify that the user can login.

Scope meaning: which parts of the system will be tested which part of the system will not be tested.

Risks: what could go wrong to the system while testing. What could go wrong to the test environment itself.

Actual answer:

Test objectives: verify that the user can login safely and can create an account if they don't have. Verify that the user can create a zoom meeting. Verify that the zoom meeting generates a url to invite others. Verify that the user can join other's sessions. Verify that the user can open their camera and mic and close them when needed.

Test Scope: signup and login modules. Meeting creation and sharing module. Accessing other services module (test camera and mic). Security measurements. Hardware; camera, mic, cpu usage, ram usage.

Risks: the system is not reliable over time and can crash during a meeting. The system is not efficient and uses too much cpu, ram. The system fails to generate a secure sharable url. The system fails to access the camera and the mic.

Test Analysis:

Task 3 – Test Analysis In your role as test manager, you are responsible for the test of the application “zoom”.

3.1.) Part of the test team is currently reviewing the test basis. Your project manager is very skeptical and cannot fully understand the benefits (compared to the effort involved). What are

the arguments in favor of your chosen test process activities?: Choosing the right test basis (requirements, user stories, existing code) helps us design efficient and complete tests. This allows us to catch defects early, reduce risks, save costs, and ensure the system meets user expectations, improving overall product quality.

3.2.) Outline the feature “Meeting participant” using at least 5 potential test conditions:

The question actually means that we need to break the feature into smaller parts that we then can test.

The user can mute and unmute the mic

The user can join a meeting

The user can open and close the camera

Multiple participants can join at the same time

The user can write in the chat

The user can leave a meeting

Select one or more test conditions and intuitively derive 5 test cases. Define preconditions, postconditions and possible test steps. What test data is required?

Test Case 1: Participant joins a meeting

Preconditions

- A meeting has been created
- Participant has a valid meeting link
- Internet connection is available

Test Steps

1. Open the meeting link
2. Enter participant name
3. Click “Join Meeting”

Postconditions

- Participant successfully joins the meeting
- Participant is visible in the participant list

Test Data

- Valid meeting URL
 - Participant name
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Test Case 2: Participant mutes and unmutes microphone

Preconditions

- Participant has joined a meeting
- Microphone is available and enabled

Test Steps

1. Click the “Mute” button
2. Try speaking
3. Click the “Unmute” button
4. Speak again

Postconditions

- Microphone is muted when muted
- Audio is restored when unmuted

Test Data

- Microphone device
 - Active meeting session
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Test Case 3: Participant turns camera on and off

Preconditions

- Participant has joined a meeting
- Camera hardware is available

Test Steps

1. Click “Start Video”
2. Verify video is visible

3. Click "Stop Video"

Postconditions

- Video is visible when enabled
- Video stops when disabled

Test Data

- Camera device
 - Active meeting
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Test Case 4: Participant sends a chat message**Preconditions**

- Participant has joined a meeting
- Chat feature is enabled

Test Steps

1. Open chat window
2. Type a message
3. Click "Send"

Postconditions

- Message is visible to other participants
- Message appears in chat history

Test Data

- Chat message text
 - Active meeting with at least one other participant
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Test Case 5: Participant leaves the meeting**Preconditions**

- Participant is currently in a meeting

Test Steps

1. Click "Leave Meeting"
2. Confirm leaving if prompted

Postconditions

- Participant is removed from the meeting
- Participant no longer appears in participant list

Test Data

- Active meeting
- Participant session